

Revised Higher Specimen: scan the code and answer these.



Q1

Q7

Q14

Section A · motion, forces, p-n

Show your working, not just the letter.

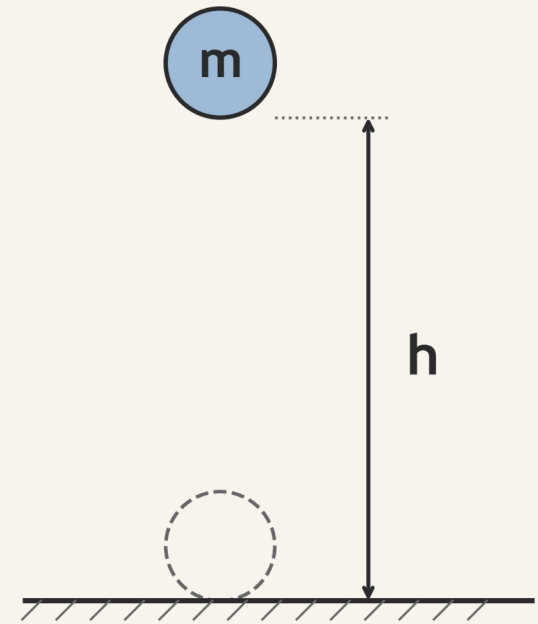


Scan for the paper
Marking scheme

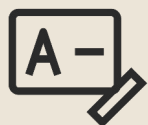
Energy and Power

Where does the energy go?

Higher Physics



What the pictures mean



Show me boards
Hold up your board.



Write it
Write it in your jotter.



Turn and talk
Talk to your partner.



Hands down
No hands up. I pick.



Hands up
Hand up to answer.



Silent task
Work alone, in silence.



Jotter notes
Copy it into your jotter.



Cut and stick
Cut out and stick in.

Two cranes lift the same load to the same height. One is twice as fast. What is different?



A tower crane

PART 1

Work done

At Higher, the force is rarely along the motion.

By the end of this part...



- 1 I can find the work done by a force at an angle
- 2 I can link work done to energy transferred, including against friction

Recap: $E_w = Fd$

$$E_w = Fd$$

E_w

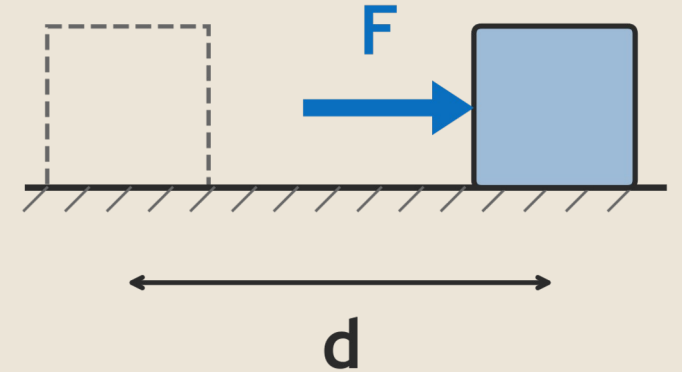
work done
(J)

F

force (N)

d

distance
(m)



Worked Example: Pulling at an Angle

A rope pulls a sledge 20 m with a 45 N force at 30° above the horizontal. Calculate the work done.

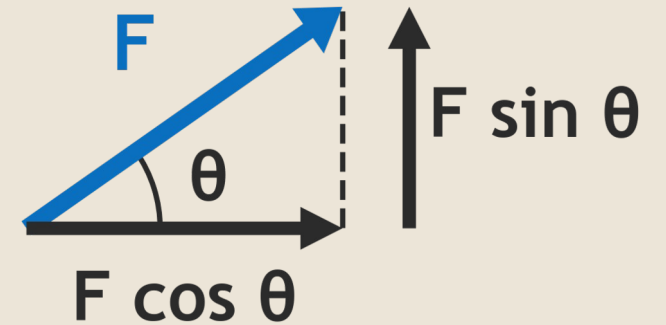
3

$$E_w = Fd \quad (1)$$

(component = $45 \cos 30^\circ = 39 \text{ N}$)

$$E_w = 39 \times 20 \quad (1)$$

$$E_w = 780 \text{ J} \quad (1)$$



Worked Example: Lifting a Load

A winch lifts a 20 kg load 3.0 m at constant speed.
Calculate the work done.

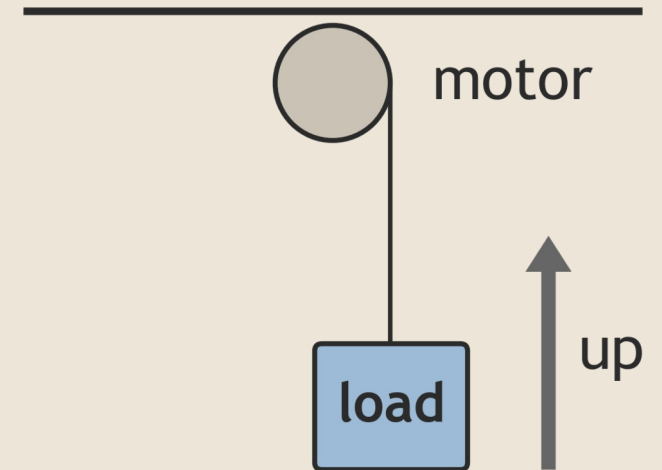
3

$$E_w = Fd \quad (1)$$

$$(F = W = mg = 20 \times 9.8 = 196 \text{ N})$$

$$E_w = 196 \times 3.0 \quad (1)$$

$$E_w = 590 \text{ J} \quad (1)$$



Your Turn: Work Done



1

A 60 N force at 20° above the horizontal pulls a cart 15 m. Calculate the work done.

850 J

2

A 12 kg box is lifted 1.5 m at constant speed. Calculate the work done.

180 J

3

Friction of 30 N acts on a sled for 25 m. Calculate the energy changed to heat.

750 J

Spot the mistake. A 40 N force pushes a box 250 cm. What is wrong with this answer?

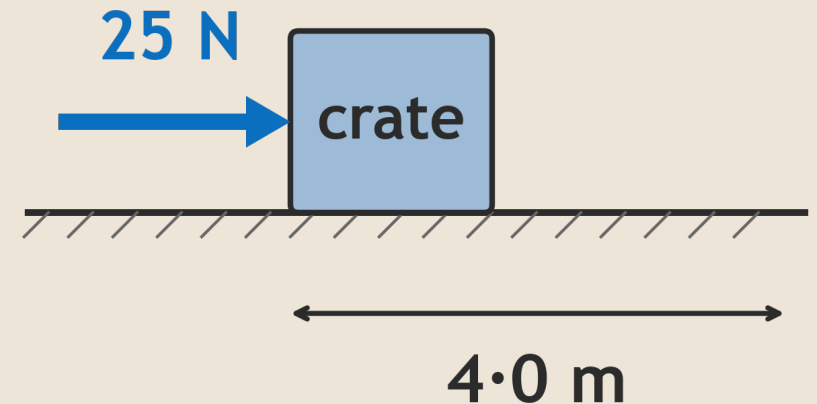


$$E_w = Fd$$

$$E_w = 40 \times 250$$

$$E_w = 10\,000\text{ J}$$

d must be in metres – so $E_w = 40 \times 2.50 = 100\text{ J}$



Plenary: Work Done



1

A force acts at an angle. Which part does work?

The component along the motion

2

Where does work against friction go?

Heat

3

A 5.0 kg box is lifted 2.0 m. Calculate E_w .

98 J

Look back — by the end of this part...



- 1 I can find the work done by a force at an angle
- 2 I can link work done to energy transferred, including against friction

2007 Higher Physics: scan the code and answer these.



Q2

Q9

Q10

Section A · motion, electricity

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 2

Energy changes

What matters is the change: in height, or in speed.

By the end of this part...



- 1 I can solve problems where E_p or E_k changes
- 2 I can use work done to find a braking force

Recap: $E_p = mgh$

$$E_p = mgh$$

E_p

gravitational
potential energy (J)

m

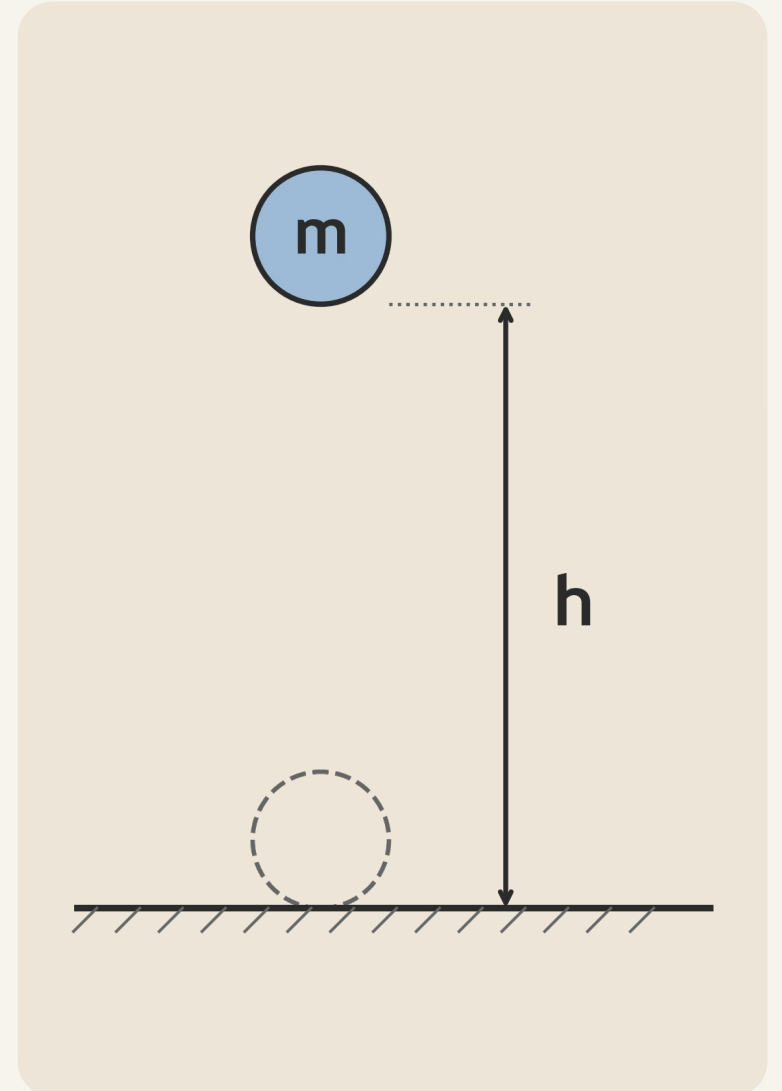
mass (kg)

g

gravitational field
strength (N kg^{-1})

h

height (m)



Recap: $E_k = \frac{1}{2}mv^2$

$$E_k = \frac{1}{2} m v^2$$

E_k

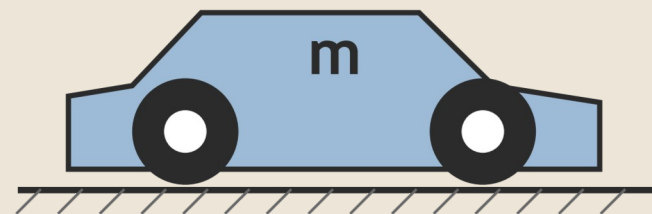
kinetic
energy (J)

m

mass (kg)

v

speed (m
 s^{-1})



Worked Example: A Crane Lift

A crane lifts a 450 kg load from 12 m to 30 m above the ground. Calculate the gain in potential energy.

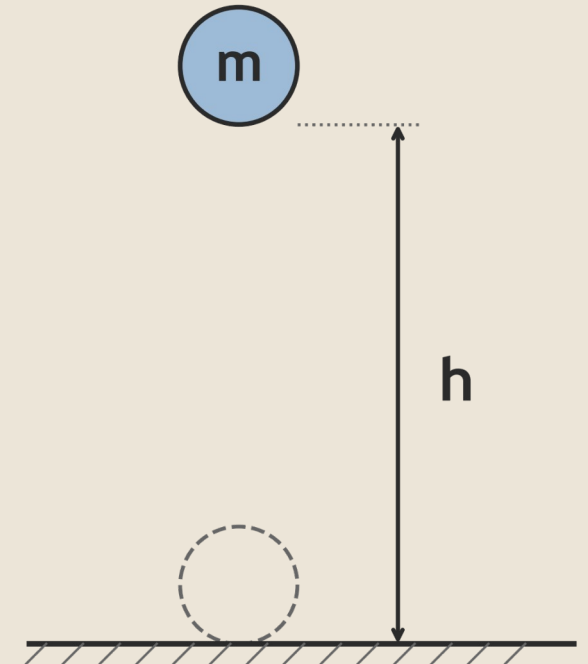
3

$$E_p = mgh \quad (1)$$

$$(h = 30 - 12 = 18 \text{ m})$$

$$E_p = 450 \times 9.8 \times 18 \quad (1)$$

$$E_p = 7.9 \times 10^4 \text{ J} \quad (1)$$



Worked Example: Speeding Up

A 1000 kg car speeds up from 10 m s^{-1} to 20 m s^{-1} .
Calculate the increase in its kinetic energy.

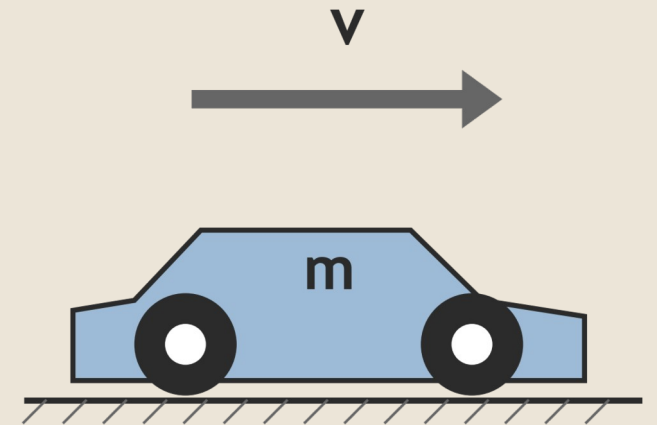
4

$$E_k = \frac{1}{2}mv^2 \quad (1)$$

$$E_k = 0.5 \times 1000 \times 10^2 = 5.0 \times 10^4 \text{ J} \quad (1)$$

$$E_k = 0.5 \times 1000 \times 20^2 = 2.0 \times 10^5 \text{ J} \quad (1)$$

$$\text{increase} = 1.5 \times 10^5 \text{ J} \quad (1)$$



Worked Example: Braking Force

A 1200 kg car at 15 m s^{-1} brakes to rest in 30 m. Calculate the average braking force.

5

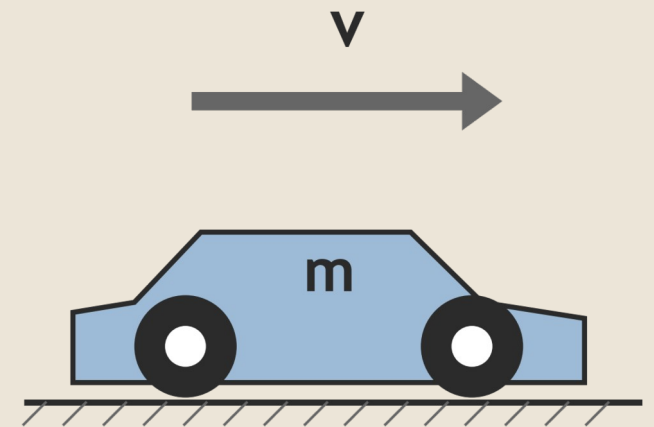
$$E_k = \frac{1}{2}mv^2 \quad (1)$$

$$E_k = 0.5 \times 1200 \times 15^2 = 1.35 \times 10^5 \text{ J} \quad (1)$$

$$E_w = Fd \quad (1)$$

$$1.35 \times 10^5 = F \times 30 \quad (1)$$

$$F = 4500 \text{ N} \quad (1)$$



Your Turn: Energy Changes



1

A 2.0 kg mass is raised from 1.5 m to 4.0 m.
Calculate its gain in E_p .

49 J

2

A 70 kg runner speeds up from 2.0 to 6.0 m s⁻¹.
Calculate their gain in E_k .

1.1×10^3 J

3

A 900 kg car at 20 m s⁻¹ brakes to rest in 40 m.
Calculate the braking force.

4500 N

Plenary: Energy Changes



1 A car's speed doubles. Same braking force: what happens to its stopping distance?

$\times 4$ — so $E_k \times 4$

2 In $E_p = mgh$, what is h ?

The change in height

3 The brakes do work equal to what?

The E_k lost

Look back — by the end of this part...



- 1 I can solve problems where E_p or E_k changes
- 2 I can use work done to find a braking force

2008 Higher Physics: scan the code and answer these.



Q1

Q2

Q3

Section A · vectors, motion, forces

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 3

Conservation of energy

Potential energy lost becomes kinetic energy gained, minus what friction takes.

By the end of this part...



1

I can use conservation of energy to solve problems with E_p and E_k

2

I can account for energy lost to friction

Recap from N5: Conservation of Energy

Energy

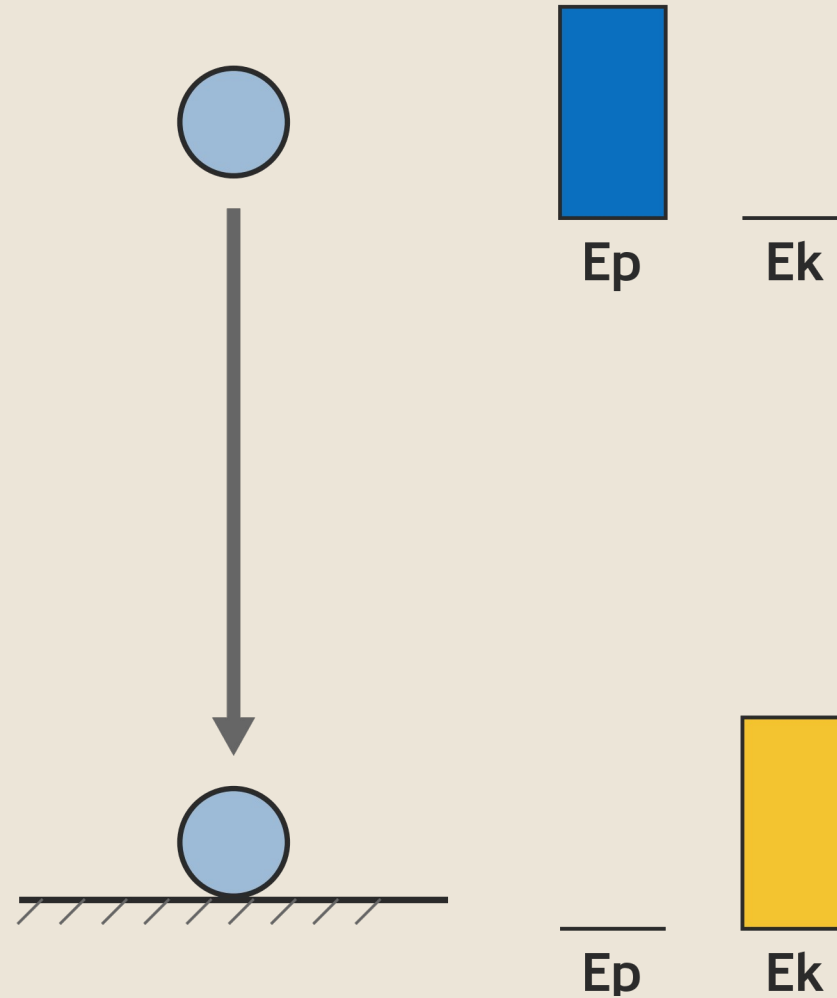
cannot be created or destroyed

No friction

$$E_p \text{ lost} = E_k \text{ gained}$$

With friction

$$E_p \text{ lost} = E_k \text{ gained} + \text{heat}$$



Worked Example: A Falling Ball

A 0.20 kg ball falls from rest through 5.0 m. Calculate its speed just before landing.

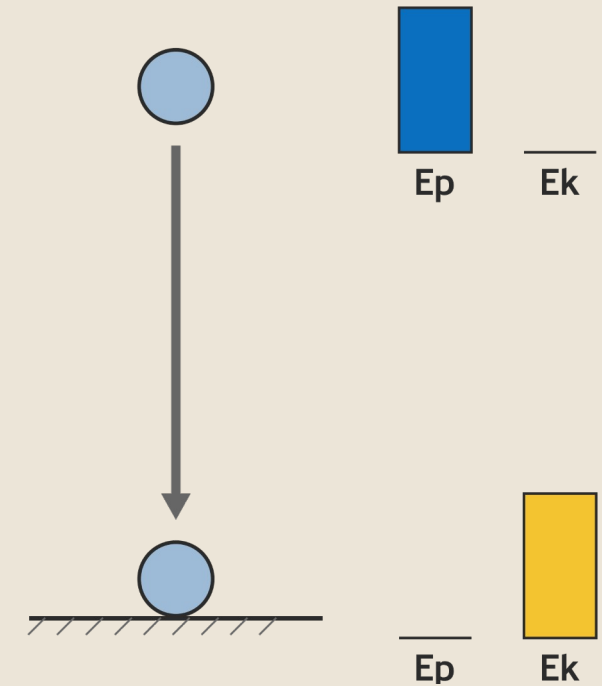
4

$$E_p = mgh \quad (1)$$

$$E_k = \frac{1}{2}mv^2 \quad (1)$$

$$0.20 \times 9.8 \times 5.0 = 0.5 \times 0.20 \times v^2 \quad (1)$$

$$v = 9.9 \text{ m s}^{-1} \quad (1)$$



Worked Example: Friction on a Carpet

A 0.25 kg toy car rolls from 0.50 m high and stops after 1.4 m of carpet. Calculate the average friction.

5

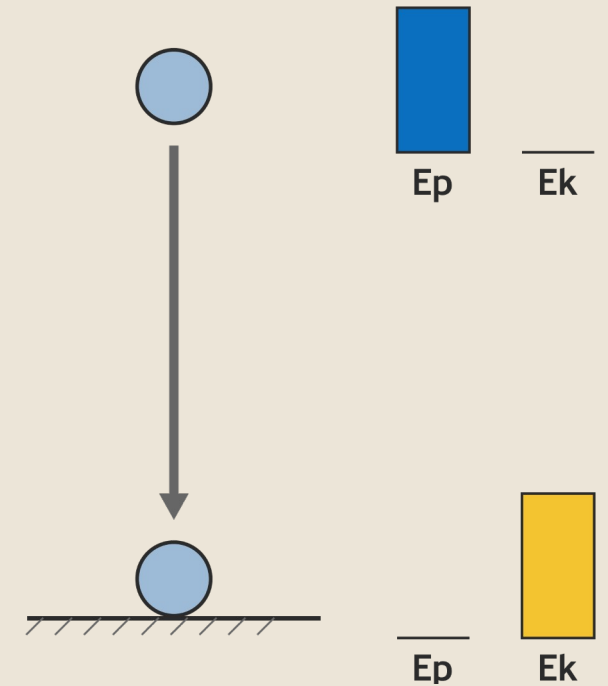
$$E_p = mgh \quad (1)$$

$$E_p = 0.25 \times 9.8 \times 0.50 = 1.2 \text{ J} \quad (1)$$

$$E_w = Fd \quad (1)$$

$$1.2 = F \times 1.4 \quad (1)$$

$$F = 0.88 \text{ N} \quad (1)$$



Your Turn: Conservation



1

A ball falls from rest through 2.0 m. Calculate its speed at the bottom.

6.3 m s^{-1}

2

A 50 kg skier drops 20 m and reaches 15 m s^{-1} . Calculate the energy lost to friction.

4200 J

3

State one assumption when E_p lost = E_k gained.

No air resistance or friction

2025 Paper 1 Q5. An object of mass 0.20 kg is released from rest 1.6 m above the ground. Its maximum speed just before it hits the ground is



A 4.0 m s⁻¹

B 5.6 m s⁻¹

C 13 m s⁻¹

D 16 m s⁻¹

E 31 m s⁻¹

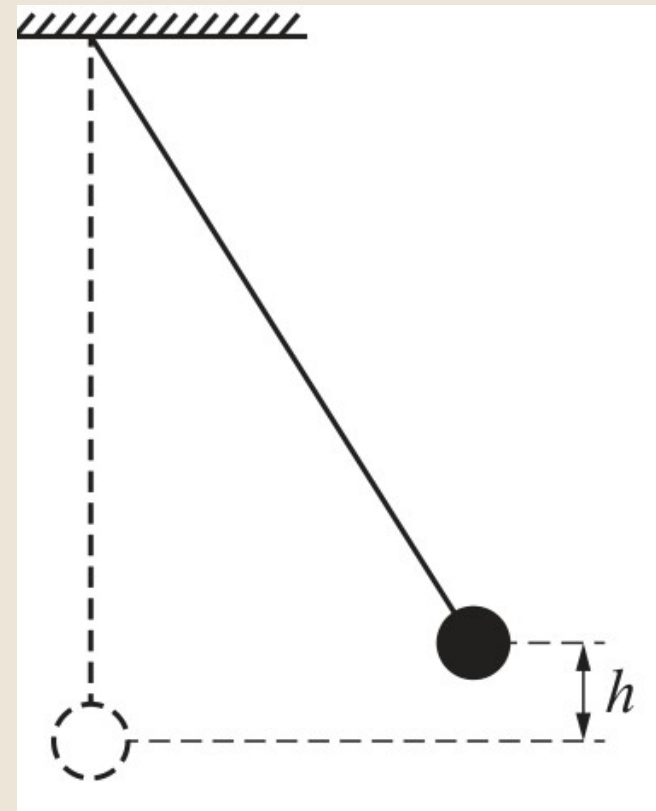
Answer B: $v^2 = 2 \times 9.8 \times 1.6$ — so $v = 5.6 \text{ m s}^{-1}$

2023 Paper 1 Q4. A pendulum bob of mass m is released from rest at height h . Neglecting air resistance, its speed at the lowest point is doubled by



- A changing the height to $4h$
- B changing the height to $2h$
- C changing the height to $h/2$
- D changing the mass to $2m$
- E changing the mass to $m/2$

Answer A: $v^2 \propto h$ — so double v needs $4h$

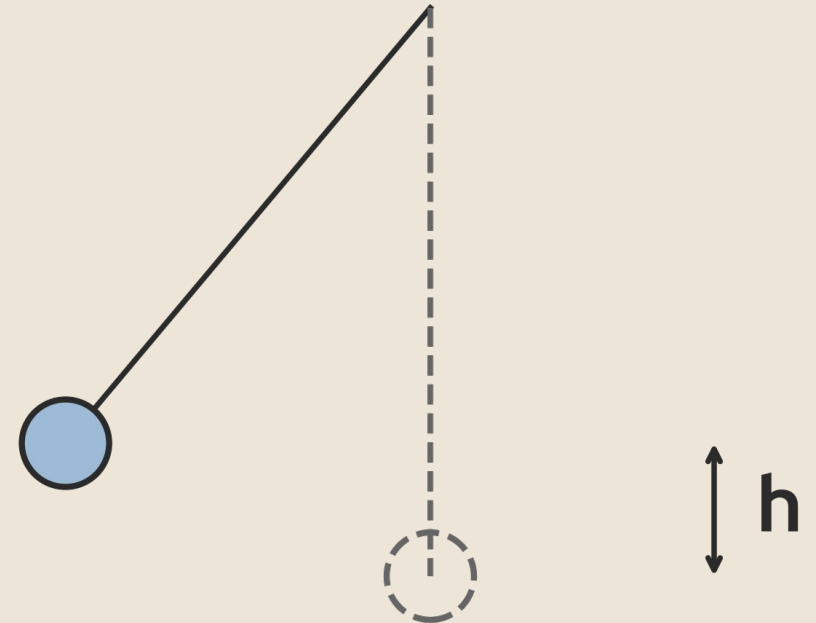


Spot the mistake. A pupil wants a ball to reach the ground twice as fast. What is wrong with their plan?



$E_p \propto h$,
so drop it from twice
the height.

$E_k \propto v^2$ — so twice the speed needs 4
times the height



Plenary: Conservation of Energy



1 With friction, E_p lost = ?

E_k gained + heat

2 Why does mass not matter for the final speed?

It cancels – so $mgh = \frac{1}{2}mv^2$

3 To double the speed, the drop height must be?

$\times 4$

Look back — by the end of this part...



1

I can use conservation of energy to solve problems with E_p and E_k

2

I can account for energy lost to friction

2011 Higher Physics: scan the code and answer these.



Q2

Q12

Q13

Section A · graphs, a.c., uncertainty

Show your working, not just the letter.



Scan for the paper

No marking scheme

PART 4

Power

The same energy, faster, is more power.

By the end of this part...



- 1 I can solve power problems that need the energy found first

Recap: $P = E / t$

$$P = E/t$$

P

power (W)

E

energy (J)

t

time (s)



Worked Example: Winching a Load

A 20 kg load is winched up 3.0 m in 4.0 s. Calculate the power.

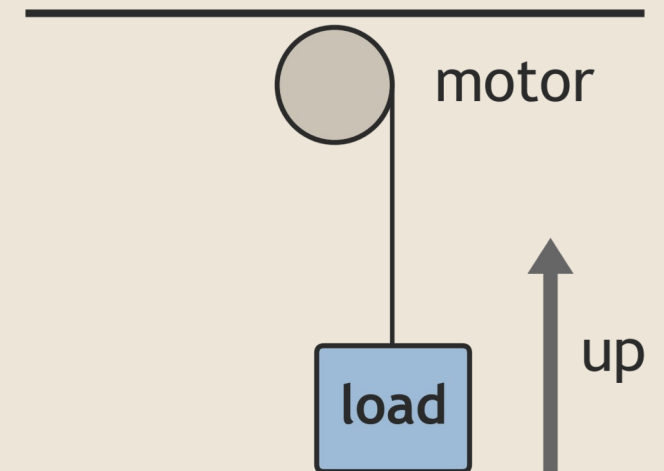
3

$$P = E/t \quad (1)$$

$$(E = mgh = 20 \times 9.8 \times 3.0 = 588 \text{ J})$$

$$P = 588 \div 4.0 \quad (1)$$

$$P = 150 \text{ W} \quad (1)$$



Worked Example: Falling Water

200 kg of water falls 50 m every second. Calculate the power delivered.

4

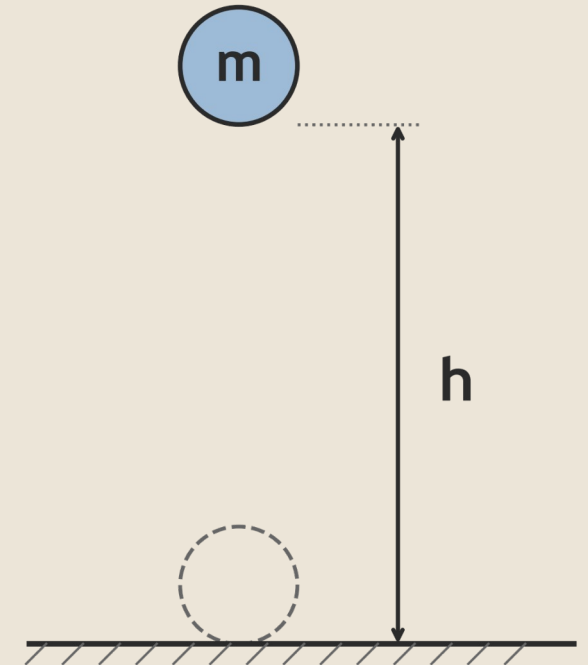
$$P = E/t \quad (1)$$

$$E = mgh = 200 \times 9.8 \times 50 \quad (1)$$

($E = 98\,000$ J each second)

$$P = 98\,000 \div 1 \quad (1)$$

$$P = 9.8 \times 10^4 \text{ W} \quad (1)$$



Your Turn: Power



1

A 1500 W motor lifts a 75 kg load 12 m. Calculate the time taken.

5.9 s

2

A pump lifts 50 kg of water 8.0 m every second. Calculate its power.

$3.9 \times 10^3 \text{ W}$

3

Two cranes lift the same load the same height. One takes half the time. Compare the power.

Double – so same energy, half the time

2022 Paper 1 Q6. Water flows at 1.0×10^6 kg per second over the Victoria Falls, which are 120 m high. The total power delivered by the falling water is:



A 1.2×10^{12} W

B 1.2×10^9 W

C 1.2×10^8 W

D 8.5×10^{-10} W

E 8.5×10^{-11} W

Answer B: $1.0 \times 10^6 \times 9.8 \times 120$ J every second – so 1.2×10^9 W

Spot the mistake. 600 kg of water flows down 20 m every minute. What is wrong with this answer?

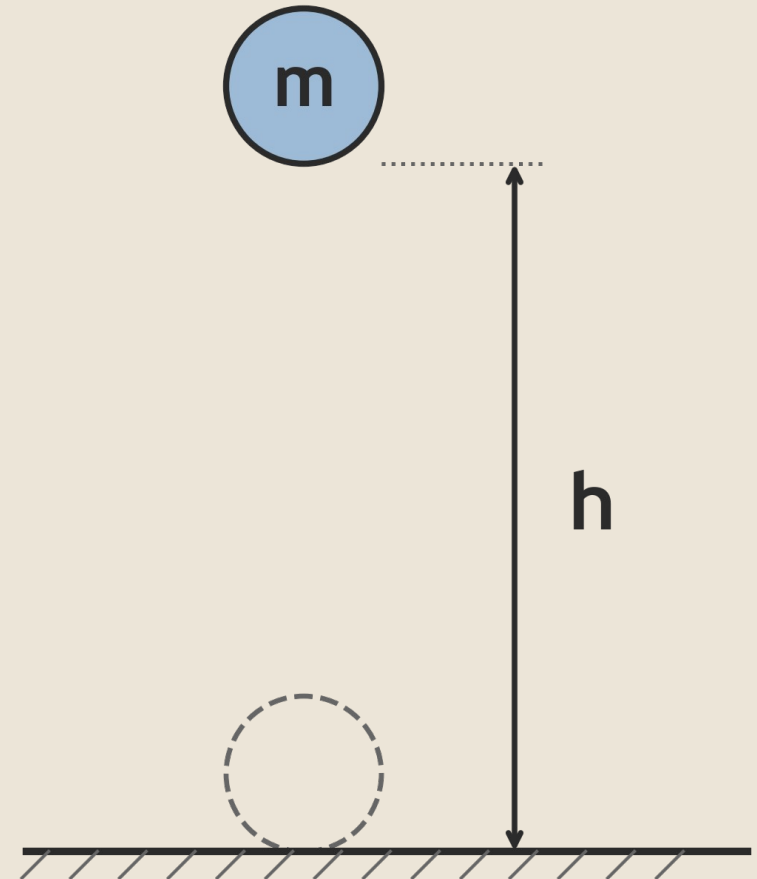


$$P = mgh$$

$$P = 600 \times 9.8 \times 20$$

$$P = 1.2 \times 10^5 \text{ W}$$

That is the energy per minute – **so $\div$ 60: $P = 2.0 \times 10^3 \text{ W}$**



Plenary: Power



1 State what 1 watt means.

1 joule per second

2 Same energy, half the time: what happens to P ?

It doubles

3 Water falls every minute. What must you do?

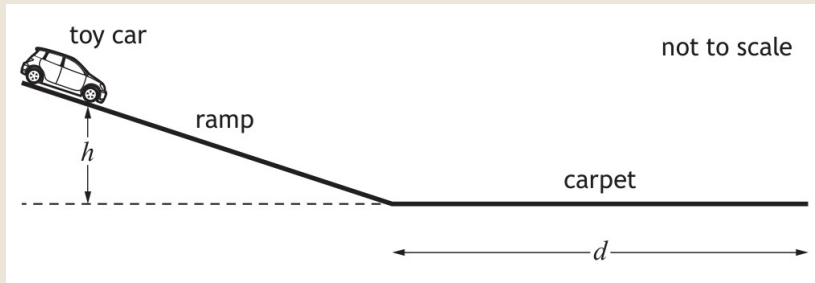
Divide by 60

Look back — by the end of this part...



- 1 I can solve power problems that need the energy found first

Energy and Power in the Exam



2016 Section 2 Q2



Past papers

Also: 2024 P1
Q5 and 2015 S1
Q6 (power from
falling water).

Brain Dump



In your jotter

Write everything you remember about energy and power.
On your own, 2 minutes. No notes.
Then check it against the plenaries.

Extension

EXTRA PRACTICE



1

A car's engine has constant power. Explain why it speeds up less as it goes faster.

$P = Fv$, so F falls — so a falls

2

Two skiers, one heavier, start from rest on the same smooth slope. Compare their speeds at the bottom.

The same — so mass cancels

LOOKING AHEAD

Next: Momentum and Collisions

What happens when moving objects meet?

Course specification

- Use of the principle of conservation of energy and appropriate relationships to solve problems involving work done, potential energy, kinetic energy, and power.

The forces bullets are deck 2a.

Credits (1 of 2)

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2023 Paper 1 Q4: © Qualifications Scotland (SQA). Higher Physics 2023, Paper 1, Question 4.

Recap: $P = E / t$: © Richard Sutcliffe, CC BY-SA 2.0, via Geograph

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Energy and Power in the Exam: © Qualifications Scotland (SQA). Higher Physics 2016, Section 2, Question 2.

Credits (2 of 2)

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v3.0, p.7